Grounwater

Depth-to-groundwater data are available on WaterNSW (in csv format) for monitoring sites across New South Wales. This assignment focuses on analysing groundwater level changes and calculating effective porosity using a combination of satellite data and soil moisture modeling.

Data Access Instructions

Depth-to-groundwater data are available from the WaterNSW realtime data website.

Data retrieval steps:

- In the "Groundwater (Telemetered data) section on the left, select "Search".
- Enter the site code in Table 1, then search for the data.
- Choose "download" found in the "Output" options. Select the sampling interval that you require.

Monitoring Site Information

Location	Site Code	Latitude	Longitude
Toganmain	GW036275	-34.6736	145.6233
Amberdell	GW036582	-35.8116	144.8309
Barratta	GW036871	-35.2736	144.5336
Howlong	GW273166	-35.9753	146.6173
Wagga Wagga	GW273167	-35.1006	147.3692
Narranderra	GW040956	-34.6966	146.3422
Jerilderie	GW036211	-35.06580	145.3797
McLellons Bore Rd	GW403747	-35.0293	145.5936
Forbes	GW036597	-33.7657	147.4806
Roeta Rd	GW036284	-33.3537	145.0300

Table 1

Questions

1. 📈 **(2 marks)** Plot the time series of depth to groundwater level for each site. Create individual plots showing the temporal variation in groundwater depth for all monitoring locations.

2. 🛰️ **(5 marks)** Calculate the groundwater change at each of the sites from a combination of satellite data and soil moisture modelling data. Use the results from previous assignment. Consider the spatial resolution limitations and interpolation methods required.

3. 💻 **(5 marks)** Estimate the **specific yield** (approximating effective porosity) at each location using (1).

$$S_y \approx n_e = \frac{\Delta S}{\Delta h} \tag{1}$$

ΔS is the change in subsurface water storage (from satellite/soil moisture data) and Δh is the observed change in groundwater level.

4. 🌍 **(5 marks)** Produce a spatial map of specific yield over the region (i.e. interpolate between the discrete data points). Use appropriate interpolation techniques to create a continuous surface map showing spatial variation in specific yield across the study region.

Important Considerations:

- **Spatial Resolution**: Find out about the optimal spatial resolution for the GRACE data. Discuss how this affects point-scale comparisons.
- **Temporal Alignment**: Ensure satellite and ground data are properly synchronized. (Select the groundwater data from the same time period as the GRACE data.)
- **Units**: Be careful with unit conversions between depth measurements (m) and storage changes (mm equivalent water depth).

Guidelines

Submission Instructions:

- Submit your answers through Canvas portal
- Include your **Python notebook** (file.ipynb or file.py)
- Submit the **groundwater depth data files** that you downloaded
- Provide a brief **methodology summary** explaining your approach to spatial interpolation and uncertainty assessment